Events and Argument Structure

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Abstract

Tracing the logical form tradition in event semantics from Davidson through Schein and Schwarzschild reveals an architecture of striking structural depth: every participant inhabits its own event layer, reference to individuals drops out entirely, and the recurring monadic, conjunctive format of these logical forms is forced by the compositional system itself á la Pietroski. Independent evidence from psycholinguistics, neurolinguistics, and event cognition converges on structurally similar conclusions. Surveying the cognitive evidence reveals a striking dissociation: evidence for prelinguistic, *polyadic* event concepts in contrast to the massively *monadic* expressions described by the event semantics surveyed here. I argue that such a dissociation can be explained by a two-level cognitive architecture on which stable compositional symbols provided by the language system are satisfied by rich, domain-specific conceptual content provided by extralinguistic cognitive system. Such moves may be leveraged to answer questions the field has long left open, such as the debate between absolute vs specific thematic roles. Together, these support reading the logical form tradition as specifying a formal, empirically-motivated candidate for a Language of Thought, specialized in representing what happens, who is involved, and how.

1 Introduction

The logical form tradition in event semantics starts with an observation about inference: *Jones buttered the toast in the kitchen* entails *Jones buttered the toast* [Davidson, 1967], a pattern that recurs as adverbials are added. Capturing such inferences forced the introduction of hidden structure, with a distinctive commitment: the internal structure of a sentence *mirrors* the structure of what it describes: its logical form corresponds to components of what happened, who was involved, and how. At the level of logical form, truth-conditional equivalence isn't the end—the structures themselves are empirical hypotheses, testable against inferential, psycholinguistic, and developmental data.

Sixty years on, that single hidden event variable has been elaborated into an architecture in which each participant inhabits its own event layer [Schein, 2017] and reference to individuals drops out entirely [Schwarzschild, 2025]—far more structure than typically supposed. Events and argument structure have been approached from many directions (see surveys in Truswell 2019; Tenny and Pustejovsky 2000; Levin and Rappaport Hovav 2019; for a philosophical treatment, Williams 2021). I take up the tradition running from Davidson [1967] through Schein and Schwarzschild, arguing that it stands in its own right as a linguistic theory of event structure, while having the most untapped potential for developing a formal architecture of the linguistic-extralinguistic boundary in cognition.

Pietroski [2018] recasts the findings of the tradition as revealing a highly-restricted format for concept composition in the mind. This connects the logical form tradition directly to the Language of Thought hypothesis—the view that high-level cognition traffics in structured, compositional representations with predicate-argument syntax—presently enjoying a reemergence in cognitive science [Quilty-Dunn et al., 2023, Mandelbaum et al., 2022]. The logical form tradition has been characterizing the Language of Thought without always naming it. Closing the loop requires a two-level architecture—stable compositional symbols satisfied by rich, domain-specific conceptual content—which also resolves a dissociation between polyadic prelinguistic event concepts, and the monadic format of logical form.

1.1 The tradition in brief

How many events does a simple transitive clause represent, and how do they relate to the named participants? Prior to Davidson [1967], sentences expressed relations between individuals with no explicit reference to events. Davidson adds a hidden, existentially quantified event variable that turns adverb-drop entailments into logical validities. Thomson [1971] lifts the count to two or three: a killing includes an action and the death that it has as a causal consequence [Pietroski, 1998, 2005]. Castañeda [1967] and Parsons [1990] separate thematic roles from verb meaning, so each participant is introduced by its own conjunct—the thesis of *total separation* (see Williams 2015; cf. partial separation, Kratzer 1996, 2003). Schein [2017] and Schwarzschild [2025] push the architecture much further (§3); Pietroski [2018] asks what cognitive system could produce such forms (§4).

"Argument structure" typically refers to thematic roles and linking. Whether thematic roles are primitives [Fillmore, 1968] or relativized clusters [Dowty, 1989, 1991], and how they map to syntax [Grimshaw, 1990, Rappaport Hovav and Levin, 1998, 2010], remains contested. In the tradition reviewed here, these questions are alive, but argument structure has a richer tapestry: it includes how participants relate to each other *via* events, how nominals contribute event-modifying content, and how "braids" of event descriptions are held together by structural relations at logical form. This richer conception is forced by entailment patterns that simpler logical forms struggle with.

1.2 What this paper does

Section 2 establishes core conceptual and logical tools built up in the logical form tradition. Section 3 presents two lesser-known advances: Schein's supermonadic architecture, which gives events to every participant, and Schwarzschild's pure event semantics, which eliminates reference to individuals. Section 4 interfaces the resulting architecture with cognition: beginning with Pietroski's view that the monadic, conjunctive format common to neo-Davidsonian event semantics is forced compositionally, I present converging cognitive and neural evidence for the architecture, sketch a two-level architecture on which the monadic logical forms are *satisfied by* the richer event representations supplied by core cognition, and read the tradition's logical forms as characterizing expressions in the Language of Thought. Section 5 takes stock and identifies open questions. In presenting this literature as components nearing a single, unified picture, I take liberties that a direct exposition would not; a full development is in progress as *The Grammar of Events*.

2 Foundations

This section traces the conceptual and logical tools built up in the logical form tradition that support the frontier advances of §3 and §4. The narrative is organized around three developments: the event argument itself, reference to events, and their internal structure.

2.1 The event argument

2.1.1 From zero events to one

Consider the inference from *John buttered the toast in the kitchen at midnight* to *John buttered the toast in the kitchen* to *John buttered the toast*. Each step drops a modifier while preserving an entailment, (1). This is puzzling on the traditional conception of the mapping from sentences to logical form: the number of arguments to the predicate appears to change with each modifier, and so no systematic logical relation can obtain between them, (2). Davidson [1967] dissolves the puzzle by postulating a hidden event argument that is existentially quantified, (3). ('$\vdash$' means 'proves'.)

(1) John buttered the toast in the kitchen. *entails*: John buttered the toast.

(2) $\text{Buttered}_3(\text{j}, \text{the toast}, \text{the kitchen}) \nvdash \text{Buttered}_2(\text{j}, \text{the toast})$

(3) $\exists e[\text{Buttered}(e, \text{j}, \text{the toast}) \mathbin{\&} \text{In}(e, \text{the kitchen})] \vdash \exists e[\text{Buttered}(e, \text{j}, \text{the toast})]$

(3) is read: There is an event e of John's buttering the toast, and e is in the kitchen. Dropping *in the kitchen* secures the inference in (3) at logical form by conjunction reduction. From the start, the event argument explains adverbial modification and variable polyadicity, while recasting a class of informal semantic observations—that *in the kitchen* characterizes the *event* of buttering, rather than, say, the toast—as consequences of logical form.

2.1.2 Separating thematic roles

How far can this architecture be pushed? Davidson's analysis preserves the traditional relation between the named arguments and the predicate. But entailment patterns reach inside the predicate, too: from *Jones buttered the toast* we can infer *Jones did something* and *Something happened to the toast*. These entailments follow if each argument is introduced by its own conjunct [Castañeda, 1967]. Parsons [1990] develops the point into a full *neo*-Davidsonian framework:

(4) $\exists e[\text{Butter}(e) \mathbin{\&} \text{Agent}(e, \text{Jones}) \mathbin{\&} \text{Patient}(e, \text{the toast}) \mathbin{\&} \text{In}(e, \text{the kitchen})]$

(5) a. $\vdash \exists e[\text{Agent}(e, \text{Jones})]$

b. $\vdash \exists e[\text{Patient}(e, \text{the toast})]$

The verb now contributes only Butter(e)—a monadic (1-place) predicate of events. Participants enter through thematic role conjuncts on a par with adverbs, and the noted entailments to individual participations are again effected by conjunction reduction.

Convergent evidence comes from nominalizations [Parsons, 1990]. The neo-Davidsonian analysis posits hidden quantification over events and silent relational expressions linking syntactic arguments to the main clause event. In nominalizations, that linking is performed overtly by prepositional material, (6b).

(6) a. The Romans destroyed the city quickly.

b. The destruction of the city by the Romans was quick.

The truth conditional equivalence of (6a) and (6b) invites an account on which one can be formally proved from the other. Vindicating that formally would require at least one more ingredient—definite reference to events (see §2.2.2).

2.2 Events as referential objects

2.2.1 Perception reports

The event argument is not only postulated for participant entailments; it can be selected by higher predicates. Perception reports provide the most direct evidence [Higginbotham, 1983]. Unlike (7), which can be true if Jones merely noted evidence confirming Mary's departure, (8) requires that Jones perceived her actual leaving:

(7) Jones saw that Mary left.

(8) Jones saw Mary leave.

Higginbotham argues that the 'naked infinitival' complement in (8) is systematically distinguished from the tensed clause in (7): it is veridical, referentially transparent, and allows quantifier export—properties of event descriptions rather than propositional complements.

On Higginbotham's account, (8) says there was a seeing event e_1 whose 'Theme' (a rather underspecified, object-relevant thematic relation) is itself an event e_2 of Mary leaving:

(9) $\exists e_1[\text{See}(e_1)\ \&\ \text{Agent}(e_1, \text{Jones})\ \&$
$\quad\quad \exists e_2[\text{Theme}(e_1, e_2)\ \&\ \text{Leave}(e_2)\ \&\ \text{Agent}(e_2, \text{Mary})]]$

2.2.2 Definite event descriptions

Schein [2020] argues that an analysis based on existential quantification will not work for perception reports like (8). The problem is sharp with an embedded negation:

(10) Jones saw Mary not leave.

The existential analysis has to interpret the negation low, but not as Boolean negation: the various options deliver truth conditions that are far too weak. Higginbotham therefore treats embedded *not* as an antonymic operator, deriving the (positive) predicate 'non-leaving.'

Schein argues that this analysis struggles on any of its logical, ontological, or grammatical interpretations, and touches it up with an analysis of the complement clause as a Fregean definite description. For the affirmative case, the two analyses are nearly equivalent, but the definite description earns its keep in the negative case:

(11) $\exists e_1[\text{See}(e_1)\ \&\ \text{Agent}(e_1, \text{Jones})\ \&$
$\quad\quad [\iota e_2 : \phi(e_2)][\text{Theme}(e_1, e_2)\ \&\ \neg\text{Leave}(e_2)\ \&\ \text{Agent}(e_2, \text{Mary})]]$

Here, Φ stands for a contextually supplied description identifying the relevant event—what John was (positively) tracking; this restricts the operator to describing that event, and in this case saying that *it* was seen and performed by Mary, but it wasn't a leaving event.

Two features of this move will recur throughout what follows. First, the shift from $\exists e$ to ιe is a shift in the *referential status* of the event variable: events are no longer merely quantified but picked out. Second, the descriptive content Φ is anaphoric—fixed by the discourse or the immediately embedding structure, not by the verb alone. What Φ amounts to systematically—how its content is determined, and by what mechanism—is the subject of *descriptive event pronouns*, taken up in §3.

2.3 The internal structure of events

2.3.1 Events have parts

What are these things that we quantify over and refer to using e variables? Thomson (1971) provides a foundational argument. On June 5, 1968, Sirhan shot Kennedy, and Kennedy died nearly 25 hours later. The initial sentences in (12) are both true of what happened—and anyone who saw the shooting might feel justified later in saying that they saw the killing—and yet it is nevertheless odd to say that the shooting and the killing happened at the same time, as felicity with the follow-ups imply.

(12) a. Sirhan shot Kennedy. It happened just after midnight on June 5th.
b. Sirhan killed Kennedy. #It happened just after midnight on June 5th.

If the shooting and the killing were the same event—as intuition might suggest—they would have to share all their properties. But they have different temporal profiles; so they must be distinct. Thomson concludes that the killing is a *complex* event that includes the shooting and the victim's death as mereological parts [Thomson, 1977, 1983].

2.3.2 Decomposing agenthood

Pietroski [1998] pushes the decomposition differently, using a similar argumentative strategy. On April 14, 1865, Booth pulled the trigger of his pistol; and the bullet entered Lincoln's skull. Both of the claims in (13) are true; and, intuition suggests they might be made true by the same event.

(13) a. Booth shot Lincoln with a pistol on April 14, 1865.
b. Booth pulled the trigger with his finger on April 14, 1865.

If the same event satisfies both logical forms, however, contradiction results: the shooting ended about when the bullet entered Lincoln; the pulling ended about when the bullet left the gun. The shooting has parts the pulling lacks; so, they must be distinct.

Pietroski's solution decomposes the Agent relation: to be Agent of event e is for some action you perform to causally ground e. Simplifying a little, *Booth shot Lincoln* looks like:

(14) $\exists e_2[\exists e_1[\text{Action}(e_1, \text{Booth}) \mathbin{\&} \text{Grounds}(e_1, e_2)] \mathbin{\&} \text{Shooting}(e_2) \mathbin{\&} \text{Patient}(e_2, \text{Lincoln})]$

On this analysis, our sense that the sentences in (13) describe the same event tracks the unity of *action*—Booth made only one major contribution to the causal order that day—while the sentences describe the variety of *consequences* that follow.

A structurally similar move by Kratzer (1996; see also 2003) proposes that the external argument is introduced by a separate functional head, building on Marantz's (1984) foundational asymmetry between internal and external arguments, and extended cross-linguistically by Pylkkänen [2008]. However, the motivations differ: Kratzer is concerned with argument introduction; Pietroski with the structure of the event itself. (See also Williams 2005, 2009 for cross-linguistic challenges to Kratzer's treatment of Theme).

2.3.3 Plural event variables

Events come in pluralities. When the Romans destroyed the city, many individual actions were required. When Jones butters 613 pastries, many individual butterings occur. And, sentences with plural noun phrases give rise to collective and distributive readings: read collectively, (15) says that there was a single carrying event performed by the boys together; read distributively, it says there was a plurality of events, each of which consisted in a carrying event by one of the boys.

(15) The boys carried the piano upstairs.

Boolos [1984] showed that plural quantification can be handled by second-order monadic variables—variables that range over "some things" without commitment to a set of those things (cf. Russell's paradox). This posit makes the collective/distributive contrast expressible: the collective reading posits a single event with a plurality of participants, (16a); the distributive reading posits a plurality of events, one for each agent, (16b). (Simplifying individual plurality:)

(16) a. $\exists e[\text{Carry}(e)\ \&\ \text{Agent}(e, \text{the boys})\ \&\ \text{Patient}(e, \text{the piano})]$
b. $\exists E[\text{Carry}(E)\ \&\ \forall x[\text{the boys}(x) \rightarrow \exists e[Ee\ \&\ \text{Agent}(e, x)\ \&\ \text{Patient}(e, \text{the piano})]]$

Plural event variables plays a big role §3, where attention to the internal structure of event pluralities—how sub-events relate to one another and to their participants—is demanded of logical form by thorny empirical data. Indeed, Schein [1993] develops arguments from combinations of plural quantification and distributivity that total separation of thematic roles is not just convenient, but essential.

2.3.4 Role Exhaustion

One further principle will matter in what follows. Role Exhaustion (RE) is the requirement that every syntactic dependent (whether argument or adjunct) θ-related to an event e uniquely and exhaustively specifies the participants bearing θ to e (see Williams 2015 gives the most thorough synthesis of the evidence in support). This supports, in principle, replacing the relational notation $Agent(e, x)$ with functional notation, $Agent(e) = x$. In Wellwood 2022, I noticed that the felicity of even simple sentences like *Denzel ran in the hallway, in the carpark* presents an apparent violation of RE, which could be addressed by analyzing higher modifier as entering via a 'framing' event layer, rather than via the ordinary clausal events (cf. Schein 2020). Indeed, as the landscape of event description changes, Role Exhaustion may be better read as a diagnostic—it tells us whether a participant has been properly connected to an event structure—and the generalizations that it summarizes may be derived from more fundamental principles.

3 The Separated Architecture

This section presents two sets of developments that push the architecture further: Schein's separated thematic structure, in which every participant inhabits its own event layer, and Schwarzschild's pure event semantics, which eliminates individual variables entirely.

3.1 Separated thematic structure

Neo-Davidsonian separation gives each participant its own conjunct but keeps them all tied to a single event variable. Schein [2017] argues that three independent pressures from natural language force pulling them apart.

3.1.1 Contrary adverbs force separation

Consider (17).

(17) Jones gracefully and Jeeves clumsily buttered 613 pastries for brunch.

The sentence entails that Jones's contribution was graceful, that Jeeves's was clumsy, and that 613 pastries got buttered. It does *not* entail that any single event was both graceful and clumsy. But any logical form that routes both agents through a single event variable—whether via some form of plural subject (e.g. $\mathrm{Agent}(E, \mathrm{Jones} + \mathrm{Jeeves})$), or by repeating the Agent role (violating Role Exhaustion)—will attribute contradictory properties to E. The fix is to give Jones and Jeeves different event variables:

(18) a. $\exists E_1\, \mathrm{W}[E_1, \mathrm{Jones}]\ \&\ \mathrm{Graceful}(E_1)$ *Jones participated, gracefully;*

b. $\&\ \ \exists E_2\, \mathrm{W}[E_2, \mathrm{Jeeves}]\ \&\ \mathrm{Clumsy}(E_2)$... *Jeeves participated, clumsily...*

The pattern generalizes to temporal and locative adverbs as well: *Saul from early morning in New Jersey and David later in the day in California have been swapping wine futures* does not locate a single event both in the morning in NJ and in the afternoon in CA. With n participants whose contributions are independently modifiable, an adequate logical form requires at least n event variables.

This does not yet specify what is the top-level relation between the participants and their events. The placeholder W just says: *something* connects Jones to his events. More can be said about this, once the second pressure is introduced.

3.1.2 Role-indeterminate coordinations motivate 'Participate'

Consider (19).

(19) The Columbia students and the Harvard students surrounded the Pentagon and were crowded into the Mall.

First fact: the sentence commits no student to either venue—some Columbia students may have surrounded the Pentagon and others been crowded into the Mall, and likewise for the Harvard students. Second fact: the two conjoined predicates demand different thematic roles for their external argument: *surround* requires Agent, *be crowded* requires Theme. To bind the outer DPs into the Agent or Theme positions inside the VP would be to insist on an assignment that the sentence leaves open. A top-level relation between the outer DP and its events—one that is silent about thematic specificity, compatible with any division of participants across the inner predicates—is required.

Schein's $\mathrm{W}[E, X]$—"participate"—serves this purpose. It says only that the Xs took part in events E, without specifying the nature of the participation. Since the participation events overlap with the described surroundings and crowdings (see Schein for the formal definition of Overlap), and since the Agent and Theme positions are existentially closed within the predicative coordination, beyond the reach of the outer quantifiers—the connection between the students and the venues is recovered inferentially.

3.1.3 Aggressor–victim asymmetry motivates the cause/effect decomposition

Underspecified participation is necessary, but not safe. Consider (20):

(20) The Harvard students and the Columbia students teased the Columbia students.

Imagine a scenario that intuitively makes (20) false: Harvard students teased Columbia students mercilessly, with Columbia students declining to tease anyone. A flat logical form that simply asserts that Harvard and Columbia participated in the teasing is accidentally true in that scenario—because Columbia *did* participate, but just as victims.

The resolution was anticipated in §2.3.2: even simple transitives decompose an agentive causing from its causal consequences. With that decomposition in hand, the teasing case resolves cleanly: Harvard students and Columbia participate in events that *cause* a being-teased, and the Columbia students participate in that being-teased. Simplifying:

(21) a. $\exists E_1\, \mathrm{W}[E_1, \mathrm{Harvard}]\ \&\ \exists E_2\, \mathrm{W}[E_2, \mathrm{Jeeves}]$ *Hs and Cs participated in events*
b. $\&\ \ \mathrm{Cause}[E_1 + E_2, E_3]$ *and those events caused*
c. $\&\ \ \exists E_3\, \mathrm{Teased}(E_3)\ \&\ \mathrm{Patient}[E_3, \mathrm{Columbia}]$ *a being-teased of Cs*

The structural position of each outer DP's events relative to the bridge relation Cause—not a thematic label—determines who counts as causer and who as victim. The Pietroski move from §2.3.2 is load-bearing for Schein's architecture.

3.1.4 Supermonadicity and bridge relations

The three pressures converge on a single architectural result. Contrary adverbs force each participant into its own event variable. Role-indeterminate coordinations force the top-level participation relation to be underspecified. The aggressor–victim asymmetry—once participation is vague—forces a cause/effect decomposition. What holds the resulting structure together is no longer a shared event variable, but *inter-event bridge relations*: Cause for causal chains, Overlap for spatial or temporal coincidence, and a small inventory of others.

Schein calls the resulting architecture *supermonadic*: distinct participants' atomic formulas do not share event variables, and all connections between participants' events are made explicit by bridge relations. This is true beyond causatives, as well: even apparently symmetric stative predicates receive supermonadic treatment. *Carnegie Hall faces Carnegie Deli* and *Carnegie Deli faces Carnegie Hall* appear to name a single static facing, but Thomson-style arguments derive contradictions from there: the Deli faces the Hall from the North, but the Hall doesn't face the Deli from the South, etc. The two facings correspond to distinct state-layers connected by topological Overlap.

3.1.5 Descriptive event pronouns

The supermonadic structure needs one further ingredient. When a bridge relation like $\mathrm{Cause}[E_1, E_2]$ picks out events introduced by earlier existentials, what exactly does it pick out? Bare existential quantification turns out to be too weak; a problem sharply visible under nonincreasing quantifiers:

(22) 18 barrels filled no more than 60 bottles.

The sentence says that *all* the filling events, collectively, involved at most 60 bottles—not that some filling events did. But $\exists E[\ldots]$ is true as long as any witness plurality exists, even when many more fillings occurred; *maximal* reference is needed. Schein [2017] proposes that the event variables in bridge relations are picked out by *descriptive event pronouns*: silent Evans-style pronouns $[\iota E : \mathrm{pro}]$ [Evans, 1980, Sharvy, 1980] whose content is anaphorically fixed by the immediately adjacent predicate structure. Under nonincreasing quantifiers, these pronouns force a maximal reading; under increasing quantifiers (*some*, *many*), maximal and existential readings converge. The locality constraint on "pro" keeps the information a speaker must track finitely specifiable and structurally bounded.

3.1.6 The full braid

We can now assemble Schein's full architecture. The canonical supermonadic logical form for (17)—adapted from Schein's Ch. 2, ex. (147)—is what he calls a *braid*: each line introduces an event layer, and backward descriptive event pronouns at the junctures refer back to events introduced earlier, their descriptive content fixed by the immediately adjacent material.

(23) Full braid: *Jones gracefully and Jeeves clumsily buttered 613 pastries for brunch*

a.	$\exists E_1\, \mathrm{W}[E_1, \mathrm{Jones}]$ & $\mathrm{Graceful}(E_1)$	*Jones participated, gracefully;*
b.	$\exists E_2\, \mathrm{W}[E_2, \mathrm{Jeeves}]$ & $\mathrm{Clumsy}(E_2)$	*Jeeves participated, clumsily;*
c.	$[\iota E_{1,2} : \mathrm{pro}_{1,2}][\iota E_3 : \mathrm{pro}_3]\, \mathrm{Cause}[E_{1,2}, E_3]$	*what they did caused*
d.	$\exists E_3\, \exists X\, \mathrm{Agent}[E_3, X]$	*some agents'*
e.	$[\iota E_3 : \mathrm{pro}_3][\iota E_4 : \mathrm{pro}_4]\, \mathrm{Cause}[E_3, E_4]$	*bringing about*
f.	$\exists E_4\, \mathrm{Butter}(E_4)$	*a buttering*
g.	$[\iota E_4 : \mathrm{pro}_4][\iota E_5 : \mathrm{pro}_5]\, \mathrm{Overlap}[E_4, E_5]$	*overlapping with*
h.	$\exists E_5\, \exists Y\, \mathrm{Patient}[E_5, Y]$	*some undergoing events*
i.	$\mathrm{Pastry}(E_5, Y)$ & $\lvert E_5 \rvert = 613$	*of 613 pastries*

Read top to bottom, with each line read as conjoined with those above it, each introduces an event layer, and each pair of adjacent layers is connected by a bridge relation whose arguments are descriptive event pronouns referring back to events just introduced. Everything that the standard neo-Davidsonian logical form packages into a single event is here distributed across five event layers, connected by Cause and Overlap, and threaded together by Evans-style descriptive event pronouns whose content is fixed by syntactic locality.

3.2 Events all the way down

Describing many more events than the neo-Davidsonian, Schein's braid still has individuals: $W[E, \mathrm{Jones}]$ names Jones as one, for example. In contrast, Schwarzschild [2025] argues that even nouns contribute descriptions of eventualities. This move completes the trajectory: the move from 0 events to many is matched by a move from many individuals to zero.

3.2.1 Nouns as state predicates

The founding move is one of symmetry. Adjectives have occasionally been analyzed as predicates of *states*, the static cousins of dynamic events; the analysis goes back at least to Higginbotham [1983] and has been pursued in varied forms since (e.g., it plays an important role in my revision of degree semantics, Wellwood 2015, 2019; Glass 2019 provides independent evidence from the English Determiner + Adjective construction). Schwarzschild reanalyzes nouns in parallel: *window* expresses a property of states of being a window, just as *open* expresses a property of states of being open. Compare the standard approach based on individuals (24), with Schwarzschild's (25).

(24) $\exists x\, [\mathrm{window}(x)$ & $\mathrm{open}(x)]$

(25) $\exists s\, \exists s'\, [\mathrm{window}(s)$ & $\mathrm{open}(s')$ & $\circledcirc(s, s')]$

(25) is read: there is a window-state s such that it has the *same participants* as an open-state s'. That both predicates fail to obtain of the same states is argued for on ontological (the identity conditions are different), grammatical (explaining the generalization that predicative adnominal adjectives are mediated by reduced relative clause structure), and inferential grounds (distinguishing direct from indirect modification). $\circledcirc$ does the work that the shared variable x did in (24): it says the same things are involved between the two predicates, but it does so without naming those things. Participants are whatever makes $\circledcirc$ true.

3.2.2 Gillon's Problem: mass nouns are singular

The sharpest pressure in favor of the symmetry move comes from a longstanding puzzle about mass nouns. Morphologically, mass nouns are singular, while semantically they seem to predicate of (anti-singular) substances or pluralities rather than objects. Evidence that mass nouns are singular comes from reciprocals, which require plural antecedents: mass descriptions pattern with singular count descriptions.

(26) a. The messages contradicted each other.
b. #The message contradicted each other.

(27) a. #The information contradicted each other.
b. # The furniture was stacked on top of each other.

Any theory that treats mass denotations as pluralities—Chierchia [1998], Landman [2000], and others—predicts that reciprocals should succeed here, but they don't. Gillon [1992] identified the issue, and Schwarzschild [2025] takes it as the forcing pressure for a new kind of denotation: mass nouns express predicates of singular states, like singular count nouns. The intuition that they are not *semantically* singular is borne out separately, next.

3.2.3 Stubbornly distributive predicates as a converging probe

A second probe confirms the picture. Some predicates—*large*, *spherical*, *long*—are "stubbornly distributive" (Schwarzschild 2025): they seem to require distribution to individuals. That they are fine with mass nouns like *furniture*, (28), creates an opening: *large* distributes over the participants of states, rather than the states themselves. Simplifying/normalizing:

(28) The furniture is large.

(29) $\exists s\,[\text{Furniture}(s)\ \&\ \exists S\,[\text{Large}(S)\ \&\ \circledcirc(s, S)]]$

Schwarzschild derives (28) as (29), with a plural use of Large: there is a single furniture-state s with participants shared, via $\circledcirc$, with a plurality S of single-participant large-states—one per participant. Distribution falls out of same-participant sharing, not from quantification over a plurality of furniture-states, while reciprocal failure is preserved at the top-level. Two independent probes, then—reciprocals and stubbornly distributive predicates—converge on the claim that mass nouns have participants, *without being pluralities*. This claim is unavailable in frameworks that treat nominal denotations as sets of individuals.

3.2.4 What "argument" becomes (Figure 1)

The architectural comparison with Schein is illuminating. Schein's W relates events and named individuals, with $W[E, X]$ saying that the Xs participated in the Es. Schwarzschild's $\circledcirc$ is a relation

$\exists E_1\, \exists s_J\, \mathrm{W}[E_1, s_J]\ \&\ \mathrm{Jones}(s_J)$	*something being Jones participated*
$[\iota E_1\colon \mathrm{pro}_1][\iota E_2\colon \mathrm{pro}_2]\, \mathrm{Cause}[E_1, E_2]$	*and what that did caused*
$\exists E_2\, \exists X\, \mathrm{Agent}[E_2, X]$	*some agenting*
$[\iota E_2\colon \mathrm{pro}_2][\iota E_3\colon \mathrm{pro}_3]\, \mathrm{Cause}[E_2, E_3]$	*bringing about*
$\exists E_3\, \mathrm{Butter}(E_3)$	*a buttering*
$[\iota E_3\colon \mathrm{pro}_3][\iota E_4\colon \mathrm{pro}_4]\, \mathrm{Overlap}[E_3, E_4]$	*overlapping with*
$\exists E_4\, \exists S_T\, \mathrm{Patient}[E_4, S_T]$	*some affected participations*
$\mathrm{Toast}(S_T)$	*of some toast*
$[\iota S_T\colon \mathrm{pro}_T][\iota S_B\colon \mathrm{pro}_B] \circledcirc [S_T, S_B]$	*sharing participants with*
$\exists S_B\, \mathrm{Brown}(S_B)$	*things being brown*

Figure 1: **The separated architecture: *Jones buttered the brown toast.*** The overall format is Schein's braid—a conjunction of monadic predications threaded by bridge relations Cause and Overlap, with cross-row reference via descriptive event pronouns—injected with an adaptation of Schwarzschild's pure event semantics: the nominals contribute stative predicates, and are connected to others by the relation of 'same participants', $\circledcirc$. [Schein, 2017, Ch. 2, ex. (147)] provided the template; tense and aspect are suppressed throughout.

between states alone: $\circledcirc(s, s')$ or the like says the same participants are involved in s and s', but does not name them. Participants are connected to events in Schein's picture via a variable X threaded through the braid; in Schwarzschild's, the "argument" is not a primitive linked to an event but what remains after state-descriptions have been woven together by $\circledcirc$.

Schwarzschild developed his framework motivated by phenomena in the nominal domain not treated by Schein, making the architectural convergence all the more striking. The two meet explicitly on counting phenomena: Schwarzschild [2025] observes that *She has taught over 2,058 students* (where students can be counted multiply) requires a description-sensitive theory of counting, citing Schein [2017] for the same point. They agree that counting operates over state- or event-structural descriptions rather than over individuals, as is proper: different descriptions generate different counts for the same sets of individuals.

Looking at the Schein-style braid and Schwarzschild's (25), the compositions involve many monadic conjunctions: the dyadic elements—$W[E, X]$, bridge relations like Cause and Overlap, Schwarzschild's $\circledcirc$—enter the derivation only to be absorbed, leaving monadic output. Figure 1 shows my gloss on the two programs at work on a single sentence, displaying the potential convergence between them. We next ask whether such convergence is an accident, or whether it reflects something deep about the compositional system.

4 Interfacing with Cognition

The monadic, conjunctive format the tradition converges on invites a cognitive question: accidental, or forced by the compositional system itself? Four moves follow: Pietroski's compositional answer and a closure of what his system composes; converging empirical evidence; a dissociation the picture resolves; a reading as Fodor's Language of Thought.

4.1 The compositional format

4.1.1 Meanings as instructions

Assuming neither Schein's supermonadicity nor Schwarzschild's pure event semantics, Pietroski [2005, 2018] develops a Chomskian view on which natural language meanings are instructions for how to build monadic concepts—a highly constrained format. Lexical meanings *fetch* concepts, and phrasal meanings are generated by application of just two compositional operations: *M-junction* conjoins two monadic concepts, (30), and *D-junction* combines a dyadic and amonadic one, linking one slot of the dyad to its monadic partner and existentially closing it, (31). Simplifying:

(30) $\Phi(\_) \;\&\; \Psi(\_)$

(31) $\exists y\, [\Delta(\_, y) \;\&\; \Phi(y)]$

The neo-Davidsonian thematic roles like Agent and Patient are dyadic thematic "concepts:" combined with the noun, D-junction existentially closes the individual argument slot, and M-joins predicates on top. The apparent dense monadic conjunction of (4) is output by a Pietroskian derivation in which every dyadic input is absorbed when introduced. The system is *massively monadic*—not because dyads are absent, but because they are compositionally transient. Pietroski calls the system SMPL (Simple Monadic Predicate Logic).

Pietroski is explicit about the cognitive character of the proposal. For him, meanings are not truth conditions, nor extensions, nor functions from possible worlds to truth values. They are *composable instructions*—recipes for how to access and assemble concepts of a special sort. The claim connects the formal architecture of event semantics directly to a hypothesis about what the mind computes in understanding a sentence.

4.1.2 Tractability and lexical gaps

Icard and Moss [2023] give SMPL a precise formal treatment, to stunning effect. I report just two results.

First, it is cognitively plausible: SMPL is complete, so reasoning in it is no harder than propositional logic (decidable, NP-complete), and a minimal extension—SMPL$^+$, adding a predicate-binding operator—reaches first-order expressive power while remaining context-free generable. The system is powerful enough to express what natural language expresses, tractable enough for fast computation, and structured enough that a finite learner could in principle acquire it. Second, certain concepts are provably inexpressible: the missing 'some not' from the traditional square of opposition is inexpressible in SMPL (cf. Horn 1972). If SMPL is the format of natural language concepts, it predicts that no natural language has a simple lexical item meaning *some_not*.

4.1.3 Two kinds of concepts

Pietroski's system stops short of specifying once and for all what are the concepts its rules fetch and combine, especially as set apart from prelinguistic concepts. He entertains [Pietroski, 2010, 2018] that lexicalization introduces concepts whose format departs from what is available prelinguistically—his "conceptual reformatting"—but does not commit to a picture of what the resulting concepts are like, or to an architecture in which the two formats stably coexist.

In ongoing work, I propose a two-level architecture. The concepts that M- and D-junction compose are what I call *G-concepts*: stable, domain-general, and ultimately higher-order compositional symbols whose type-theoretic structure lets them combine freely. *S-concepts*, in contrast, are rich, domain-specific representations in special-purpose cognitive systems that categorize events, their participants, and other things. G-concepts are *satisfied by* S-concepts [Wellwood, submitted]: Pietroski's rules describe how G-concepts compose, and the compositional format they identify is the format of G-concepts.

Together with Schein and Schwarzschild, this is an argument that the logical forms of natural language are far from notational conveniences: they present precise hypotheses about what we understand when we understand the meaning of a sentence—about the concepts the language faculty builds and how they hook into the cognitive systems that supply their anchoring to the world. If this picture is right, the dimensions the separated architecture encodes should be ones the mind tracks independently—and they are.

4.2 Independent convergence

4.2.1 Thematic computation in real time

Thematic roles are computed incrementally and predictively. Altmann and Kamide [1999] found that listeners launch anticipatory eye movements to thematically appropriate referents before they are mentioned. Kamide et al. [2003] showed that Agent plus Verb jointly anticipate Patient. And Trueswell et al. [1994] demonstrated that thematic fit resolves syntactic ambiguity online. These computations are thought to occur over rich, verb-specific event representations [McRae et al., 1997].

Verb-specificity has long unsettled the classical logical form tradition, which Dowty [1991] diagnoses as a standoff between absolute thematic categories and relativized proto-role entailments—both of which generate familiar problems (cf. Rissman and Rawlins 2017). The two-level view of §4.1.3 relocates the generalization: unity lives at the G-conceptual level (AGENT, PATIENT), variety at the S-conceptual content that satisfies it. The psycholinguistic data sit naturally on this picture.

4.2.2 Neural evidence

Neural evidence tracks the same distinction between structural role-computation and lexical-semantic content. Bornkessel and Schlesewsky [2006], Bornkessel-Schlesewsky and Schlesewsky [2016] developed a neurocognitive model organized around phases of argument-structure computation, arrived at independently of formal semantics. Kuperberg et al. [2020] show that thematic-role violations involving animacy elicit an enhanced P600, while lexical-semantic violations elicit N400 effects [Kuperberg, 2016]—suggesting that argument roles are computed at a level distinct from lexical semantics. Blackett et al. [2022] confirm with lesion data that thematic semantic errors in aphasia implicate a dissociable anteroposterior temporal network. And Ünal et al. [2024] show that salience asymmetries among Agents, Patients, Goals, and Instruments pattern similarly across linguistic and visual-search tasks.

4.2.3 Event cognition and segmentation

A second body of evidence converges from studies of event perception and event cognition. Event Segmentation Theory [Zacks and Tversky, 2001, Radvansky and Zacks, 2014] holds that the mind maintains active event models updated at boundaries driven by changes in agent, time, space, causation, and goal—dimensions closely parallel to those the separated architecture encodes as distinct event layers. Kumar et al. [2023] connect formal probability to event-model updating, showing that Bayesian surprise predicts where listeners mark event boundaries. In discourse, Zwaan and Radvansky [1998] argue for multi-dimensional situation models tracking the same dimensions.

Cross-linguistic work on how events are perceived, bounded, and lexicalized [Ji and Papafragou, 2018, Lee and Papafragou, 2024] finds event roles and event units patterning similarly across language and cognition. And Bedny et al. [2014] showed that the left middle temporal gyrus tracks event semantics, not grammatical category: event nouns like *hurricane* activate the same network as verbs, while object nouns do not. In light of Schwarzschild's work, this raises the question of an event/state split. Importantly, none of these programs was designed to test the separated architecture, and so their convergence on the dimensions it encodes is suggestive, and unlikely to be a coincidence.

4.3 A format dissociation and its resolution

4.3.1 The dissociation

Evidence suggests that human infants possess three-place event representations—distinguishing at least Agent, Patient, and Instrument—well before productive language [Gordon, 2003, Wellwood et al., 2015, Perkins et al., 2025]. But the logical form tradition gives us *monadic* event structures: Schein's supermonadicity constructs monadic predicates of events linked by bridge relations, not polyadic thematic predicates; Schwarzschild's pure event semantics eliminates individual variables entirely. If prelinguistic event concepts can be polyadic or better, and if the compositional system that generates logical forms can only compute monads, then these prelinguistic concepts are not composable by that system—the formats do not match. The two-level view of §4.1.3 resolves this tension without flattening it: the monadic LF is a G-conceptual structure, while the polyadic event representation is S-conceptual; the former is *satisfied by* the latter. This raises to the fore questions about thematic linking as an interface problem.

4.3.2 Evidence and acquisition

Evidence for such a division of labor is abundant on the S-conceptual side. Core knowledge systems provide infants with early-emerging ontological primitives—objects, agents, numbers, spatial relations [Spelke and Kinzler, 2007]—on which the language system builds [Strickland, 2017]. The agency system is especially rich [Leslie, 1995, Woodward, 1998], and preverbal infants already exhibit precursors to logical reasoning over these representations [Cesana-Arlotti et al., 2018].

Direct experimental evidence for the psychological reality of the event-ontological categories the logical form tradition trades in comes from work pursuing Bach's [1986] object/substance::event/process analogy [Wellwood et al., 2018a,b]: the cognitive representations underpinning the count/mass and atelic/telic distinctions respect the same formal distinctions; diagnostics from comparative *more* confirm the same [Wellwood, 2020]. Rissman and Majid [2019] provide cross-linguistic evidence broadly consistent with the two-level view: thematic role structure is universal for core roles like Agent and Patient, more variable for others; such patterning may be explicable in terms of the kinds of S-conceptual inventory and their recoding into G-conceptual format.

How do children bridge the two levels? Syntactic bootstrapping [Gleitman, 1990, Naigles, 1990, Fisher, 1996, Lidz et al., 2003] provides part of the answer, and Perkins et al. [2025] show for event participants in particular that thematic content is more important than the number of syntactic arguments for honing in on an event type. As Knowlton et al. [2021] articulate, linguistic meanings are cognitive instructions; on the two-level view, the instructions build G-concepts, while the S-concepts that satisfy them are often doing categorizing work before the learner encounters the word.

4.3.3 The cognitive significance of the formal apparatus

The formal apparatus of §3 now takes on specific cognitive significance. Schwarzschild's $\circledcirc$ is the mechanism by which the compositional system connects predications *without naming individuals*: two state-descriptions are linked by shared participation, not by a shared variable. The individuals themselves—tracked, categorized, and enumerated by the S-conceptual systems that core knowledge provides—are what make $\circledcirc$ true at the level of interpretation. The compositional system builds G-conceptual structures; the S-conceptual systems supply the participants that satisfy them. How much of the work Schein's iota-pronouns and descriptive anaphors do is handled in the G-conceptual layer, and where exactly S-conceptual content enters, is a question that the present synthesis raises, but cannot resolve.

4.4 The language of thought

Recent cognitive science has accumulated evidence for Language of Thought (LoT)-like structure across cognition. Quilty-Dunn et al. [2023] and Mandelbaum et al. [2022] argue that properties like role-filler independence and discrete constituent structure—hallmarks of the classical LoT hypothesis—show up across vision, motor planning, and infant reasoning. Neural evidence finds compositional representations in geometric perception [Dehaene et al., 2022, Al Roumi et al., 2021] and in predicate-argument composition during sentence processing [Frankland and Greene, 2020], and some computational models posit LoT-style primitives for learning and inference [Piantadosi et al., 2016].

Fodor's original LoT [Fodor, 1975] was stronger: a fully general, fully systematic compositional system answerable to the Generality Constraint [Evans, 1982]. I contend that the logical form tradition has been characterizing this system without always naming it. Almost to the definitional letter, the G-conceptual system can serve. The only *mis*alignment is the following: Fodor [1998] held that one and the same representational vehicle tokened in thought was tokened in categorization—that such "concepts" were the *sine qua non* of recognizing a thing as being of a certain type. The two-level view severs this: G-concepts compose sentence meanings and S-concepts recognize instances. Moreover, the compositional framework of §4.1 is, on my read, G-conceptual combinatorics: a cognitive hypothesis about the system whose outputs are the logical forms the tradition has built.

The upshot is that the separated architecture is not merely a notational scheme for predicting entailment patterns. It is a hypothesis about the format of the representations the language faculty computes—representations primarily motivated through extended consideration of linguistic examples, independently motivated by their tractability [Icard and Moss, 2023] and their convergence with structures from psycholinguistics and cognitive neuroscience, and their promise for interfacing, via the G/S-concept distinction, with the rich event representations provided by other systems of mind.

5 Conclusion

The logical form tradition's distinctive contribution is the thesis that a sentence's form mirrors the structure of what it describes, and that this mirroring is detectable empirically. Schein's separated architecture threads each participant through its own event layer; Schwarzschild's pure event semantics removes individual variables entirely; and Pietroski's M-/D-junction calculus identifies a minimal compositional system to generate the resultant forms. A two-level architecture completes the picture: the tradition's logical forms constitute G-concepts, higher-order representations satisfied by S-concepts that anchor them to the world. The resulting monadic, conjunctive logical forms constitute testable hypotheses about what the language faculty computes, independently supported by cognitive and neural evidence that was not designed to test them.

The open questions are best read as stakes for the program. If the separated architecture provides appropriate characterizations of semantic competence, it should extend to attitudes and modals—core empirical arenas better covered in other traditions; important groundwork has been laid [Hacquard, 2006, 2010, Kratzer, 2006, Moulton, 2009, Wurmbrand and Lohninger, 2023, Cariani et al., 2024], and failure would be informative—we might find evidence that different structural constraints govern different grammatical domains. Whether statives require their own variable type [Kim, 1976, Maienborn, 2005] bears on whether the G-conceptual system is unified or domain-split. And rival architectures—force-dynamic [Talmy, 1988, Copley and Harley, 2015] and truthmaker [Fine, 2017, Champollion, 2025, Bernard and Champollion, 2024]—might challenge whether the event variable is primitive at all. Articulating the mature logical form tradition precisely enough to make the comparisons sharp is part of the point of this review.

The upshot is a research agenda. The G/S-concept distinction makes testable predictions about language processing, acquisition, and breakdown: where monadic format constraints should show up, and where the polyadic content of core cognition should matter. The convergence between formal semantics and event cognition is now tight enough that the two traditions need not continue in parallel: the logical form tradition has been building what looks increasingly like the architecture that the reemerging Language of Thought program needs. In my view, the next decade of work on events and argument structure will be most productive where these traditions meet.

References

Fosca Al Roumi, Sébastien Marti, Liping Wang, Marie Amalric, and Stanislas Dehaene. Mental compression of spatial sequences in human working memory using numerical and geometrical primitives. *Neuron*, 109(16):2627–2639, 2021.

Gerry T. M. Altmann and Yuki Kamide. Incremental interpretation at verbs: Restricting the domain of subsequent reference. *Cognition*, 73(3):247–264, 1999. doi: 10.1016/S0010-0277(99)00059-1.

Emmon Bach. The algebra of events. *Linguistics and Philosophy*, 9(1):5–16, 1986.

Marina Bedny, Swethasri Dravida, and Rebecca Saxe. Shindigs, brunches, and rodeos: The neural basis of event words. *Cognitive, Affective, & Behavioral Neuroscience*, 14(3):891–901, 2014. doi: 10.3758/s13415-013-0217-z.

Timothée Bernard and Lucas Champollion. Negative events and compositional semantics. *Journal of Semantics*, 40(4):585–637, 2024. doi: 10.1093/jos/ffad018.

Deena Schwen Blackett, Jesse Varkey, Janina Wilmskoetter, Rebecca Roth, Julius Fridriksson, Gregory Hickok, and Leonardo Bonilha. Neural network bases of thematic semantic processing in language production. *Cortex*, 156:126–143, 2022. doi: 10.1016/j.cortex.2022.08.007.

George Boolos. To be is to be the value of a variable (or some values of some variables). *Journal of Philosophy*, 81:430–450, 1984.

Ina Bornkessel and Matthias Schlesewsky. The extended argument dependency model: A neurocognitive approach to sentence comprehension across languages. *Psychological Review*, 113(4): 787–821, 2006. doi: 10.1037/0033-295X.113.4.787.

Ina Bornkessel-Schlesewsky and Matthias Schlesewsky. The argument dependency model. In Gregory Hickok and Steven L. Small, editors, *Neurobiology of Language*, pages 401–414. Academic Press, 2016. doi: 10.1016/B978-0-12-407794-2.00030-4.

Fabrizio Cariani, Paolo Santorio, and Alexis Wellwood. Confidence reports. *Semantics and Pragmatics*, 2024.

Hector-Neri Castañeda. Comments on Donald Davidson 'The logical form of action sentences'. In Nicholas Rescher, editor, *The Logic of Decision and Action*, pages 104–112. Pittsburgh University Press, Pittsburgh, 1967.

Nicolò Cesana-Arlotti, Ana Martín, Ernő Téglás, Lara Voigt, Ryszard Cetnarski, and Luca L. Bonatti. Precursors of logical reasoning in preverbal human infants. *Science*, 359(6381):1263–1266, 2018.

Lucas Champollion. Truthmaker semantics and natural language semantics. *Language and Linguistics Compass*, 19(1):e70004, 2025. doi: 10.1111/lnc3.70004.

Gennaro Chierchia. Reference to kinds across languages. *Natural Language Semantics*, 6:339–405, 1998. doi: 10.1023/A:1008324218506.

Bridget Copley and Heidi Harley. A force-theoretic framework for event structure. *Linguistics and Philosophy*, 38(2):103–158, 2015. doi: 10.1007/s10988-015-9168-x.

Donald Davidson. The logical form of action sentences. In Nicholas Rescher, editor, *The Logic of Decision and Action*, pages 81–95. University of Pittsburgh Press, Pittsburgh, 1967.

Stanislas Dehaene, Fosca Al Roumi, Yair Lakretz, Samuel Planton, and Mathias Sablé-Meyer. Symbols and mental programs: A hypothesis about human singularity. *Trends in Cognitive Sciences*, 26(9):751–766, 2022.

David R. Dowty. On the semantic content of the notion of 'thematic role'. In Gennaro Chierchia, Barbara H. Partee, and Raymond Turner, editors, *Properties, Types, and Meanings, Vol. II: Semantic Issues*, pages 69–130. Kluwer, Dordrecht, 1989. doi: 10.1007/978-94-009-2723-0_3.

David R. Dowty. Thematic proto-roles and argument selection. *Language*, 67(3):547–619, 1991. doi: 10.2307/415037.

Gareth Evans. Pronouns. *Linguistic Inquiry*, 11(2):337–362, 1980.

Gareth Evans. *The Varieties of Reference*. Oxford University Press, Oxford, 1982. Edited by John McDowell.

Charles J. Fillmore. The case for case. In Emmon Bach and Robert T. Harms, editors, *Universals in Linguistic Theory*, pages 1–88. Holt, Rinehart and Winston, New York, 1968.

Kit Fine. Truthmaker semantics. In Bob Hale, Crispin Wright, and Alexander Miller, editors, *A Companion to the Philosophy of Language*, pages 556–577. Wiley-Blackwell, 2nd edition, 2017.

Cynthia Fisher. Structural limits on verb mapping: The role of analogy in children's interpretations of sentences. *Cognitive Psychology*, 31(1):41–81, 1996. doi: 10.1006/cogp.1996.0012.

Jerry A. Fodor. *The Language of Thought.* Harvard University Press, Cambridge, MA, 1975.

Jerry A. Fodor. *Concepts: Where Cognitive Science Went Wrong.* Oxford University Press, Oxford, 1998. doi: 10.1093/0198236360.001.0001.

Steven M. Frankland and Joshua D. Greene. Concepts and compositionality: In search of the brain's language of thought. *Annual Review of Psychology*, 71:273–303, 2020.

Brendan S. Gillon. Towards a common semantics for English count and mass nouns. *Linguistics and Philosophy*, 15(6):597–639, 1992. doi: 10.1007/BF00628112.

Lelia Glass. Adjectives relate individuals to states: Evidence from the two readings of English Determiner + Adjective. *Glossa: A Journal of General Linguistics*, 4(1):1–32, 2019. doi: 10.5334/gjgl.552.

Lila Gleitman. The structural sources of verb meanings. *Language Acquisition*, 1(1):3–55, 1990.

Peter Gordon. The origin of argument structure in infant event representations. In *Proceedings of the 26th Annual Boston University Conference on Language Development*, Somerville, MA, 2003. Cascadilla Press.

Jane Grimshaw. *Argument Structure.* Number 18 in Linguistic Inquiry Monographs. MIT Press, Cambridge, MA, 1990.

Valentine Hacquard. *Aspects of Modality.* PhD thesis, MIT, 2006.

Valentine Hacquard. On the event relativity of modal auxiliaries. *Natural Language Semantics*, 18 (1):79–114, 2010. doi: 10.1007/s11050-010-9056-4.

James Higginbotham. The logic of perceptual reports: An extensional alternative to situation semantics. *The Journal of Philosophy*, 80(2):100–127, 1983.

Laurence R. Horn. *On the Semantic Properties of Logical Operators in English.* PhD thesis, University of California, Los Angeles, 1972.

Thomas F. Icard and Lawrence S. Moss. A simple logic of concepts. *Journal of Philosophical Logic*, 52(3):705–730, 2023. doi: 10.1007/s10992-022-09685-1.

Yue Ji and Anna Papafragou. Midpoints and endpoints in event perception. In *Proceedings of the 40th Annual Meeting of the Cognitive Science Society*, pages 1894–1899, 2018.

Yuki Kamide, Gerry T. M. Altmann, and Sarah L. Haywood. The time-course of prediction in incremental sentence processing: Evidence from anticipatory eye movements. *Journal of Memory and Language*, 49(1):133–156, 2003. doi: 10.1016/S0749-596X(03)00023-8.

Jaegwon Kim. Events as property exemplifications. In Myles Brand and Douglas Walton, editors, *Action Theory*, pages 159–177. D. Reidel, Dordrecht, 1976.

Tyler Knowlton, Tim Hunter, Darko Odic, Alexis Wellwood, Justin Halberda, Paul Pietroski, and Jeffrey Lidz. Linguistic meanings as cognitive instructions. *Annals of the New York Academy of Sciences*, 1500(1):134–144, 2021. doi: 10.1111/nyas.14618.

Angelika Kratzer. Severing the external argument from its verb. In Johan Rooryck and Laurie Zaring, editors, *Phrase Structure and the Lexicon*, pages 109–137. Kluwer, Dordrecht, 1996.

Angelika Kratzer. The event argument and the semantics of verbs. Unpublished manuscript, University of Massachusetts Amherst, 2003.

Angelika Kratzer. Decomposing attitude verbs. Talk given at The Hebrew University of Jerusalem, 2006.

Manoj Kumar, Ariel Goldstein, Sebastian Michelmann, Jeffrey M. Zacks, Uri Hasson, and Kenneth A. Norman. Bayesian surprise predicts human event segmentation in story listening. *Cognitive Science*, 47(10):e13343, 2023. doi: 10.1111/cogs.13343.

Gina R. Kuperberg. Separate streams or probabilistic inference? What the N400 can tell us about the comprehension of events. *Language, Cognition and Neuroscience*, 31(5):602–616, 2016. doi: 10.1080/23273798.2015.1130233.

Gina R. Kuperberg, Trevor Brothers, and Edward W. Wlotko. A tale of two positivities and the N400: Distinct neural signatures are evoked by confirmed and violated predictions at different levels of representation. *Journal of Cognitive Neuroscience*, 32(1):12–35, 2020. doi: 10.1162/jocn_a_01465.

Fred Landman. *Events and Plurality*. Kluwer, Dordrecht, 2000.

Sarah Hye-yeon Lee and Anna Papafragou. Event-general conceptual categories organize verb semantics and acquisition cross-linguistically. In *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*, 2024.

Alan M. Leslie. A theory of agency. In Dan Sperber, David Premack, and Ann James Premack, editors, *Causal Cognition: A Multidisciplinary Debate*, pages 121–149. Clarendon Press, Oxford, 1995.

Beth Levin and Malka Rappaport Hovav. Lexicalization patterns. In Robert Truswell, editor, *The Oxford Handbook of Event Structure*, pages 395–425. Oxford University Press, Oxford, 2019. doi: 10.1093/oxfordhb/9780199685318.e.18.

Jeffrey Lidz, Henry Gleitman, and Lila Gleitman. Understanding how input matters: Verb learning and the footprint of universal grammar. *Cognition*, 87(3):151–178, 2003. doi: 10.1016/S0010-0277(02)00230-5.

Claudia Maienborn. On the limits of the Davidsonian approach: The case of copula sentences. *Theoretical Linguistics*, 31(3):275–316, 2005. doi: 10.1515/thli.2005.31.3.275.

Eric Mandelbaum, Yarrow Dunham, Roman Feiman, Chaz Firestone, E. J. Green, Daniel Harris, Melissa M. Kibbe, Benedek Kurdi, Myrto Mylopoulos, Joshua Shepherd, Alexis Wellwood, Nicolas Porot, and Jake Quilty-Dunn. Problems and mysteries of the many languages of thought. *Cognitive Science*, 46(12):e13225, 2022. doi: 10.1111/cogs.13225.

Alec Marantz. *On the Nature of Grammatical Relations*. Linguistic Inquiry Monographs 10. MIT Press, Cambridge, MA, 1984.

Ken McRae, Todd R. Ferretti, and Liane Amyote. Thematic roles as verb-specific concepts. *Language and Cognitive Processes*, 12(2–3):137–176, 1997. doi: 10.1080/016909697386793.

Keir Moulton. *Natural Selection and the Syntax of Clausal Complementation.* PhD thesis, University of Massachusetts Amherst, 2009.

Letitia Naigles. Children use syntax to learn verb meanings. *Journal of Child Language*, 17(2): 357–374, 1990. doi: 10.1017/S0305000900013817.

Terence Parsons. *Events in the Semantics of English.* MIT Press, Cambridge, MA, 1990.

Laurel Perkins, Tyler Knowlton, Alexander Williams, and Jeffrey Lidz. Thematic content, not number matching, drives syntactic bootstrapping. *Language Learning and Development*, 2025.

Steven T. Piantadosi, Joshua B. Tenenbaum, and Noah D. Goodman. The logical primitives of thought: Empirical foundations for compositional cognitive models. *Psychological Review*, 123 (4):392–424, 2016.

Paul Pietroski. Actions, adjuncts, and agency. *Mind*, 107:73–112, 1998.

Paul M. Pietroski. *Events and Semantic Architecture.* Oxford University Press, Oxford, 2005.

Paul M. Pietroski. Concepts, meanings, and truth: First nature, second nature, and hard work. *Mind and Language*, 25(3):247–278, 2010. doi: 10.1111/j.1468-0017.2010.01390.x.

Paul M. Pietroski. *Conjoining Meanings: Semantics Without Truth Values.* Oxford University Press, Oxford, 2018.

Liina Pylkkänen. *Introducing Arguments.* MIT Press, Cambridge, MA, 2008.

Jake Quilty-Dunn, Nicolas Porot, and Eric Mandelbaum. The best game in town: The reemergence of the language-of-thought hypothesis across the cognitive sciences. *Behavioral and Brain Sciences*, 46:e261, 2023. doi: 10.1017/S0140525X22002849.

Gabriel A. Radvansky and Jeffrey M. Zacks. Event cognition. *Psychological Bulletin*, 140(5): 1295–1327, 2014. doi: 10.1037/a0036664.

Malka Rappaport Hovav and Beth Levin. Building verb meanings. In Miriam Butt and Wilhelm Geuder, editors, *The Projection of Arguments: Lexical and Compositional Factors*, pages 97–134. CSLI Publications, Stanford, CA, 1998.

Malka Rappaport Hovav and Beth Levin. Reflections on manner/result complementarity. In Malka Rappaport Hovav, Edit Doron, and Ivy Sichel, editors, *Lexical Semantics, Syntax, and Event Structure*, pages 21–38. Oxford University Press, Oxford, 2010. doi: 10.1093/acprof:oso/9780199544325.003.0002.

Lilia Rissman and Asifa Majid. Thematic roles: Core knowledge or linguistic construct? *Psychonomic Bulletin & Review*, 26(6):1850–1869, 2019. doi: 10.3758/s13423-019-01634-5.

Lilia Rissman and Kyle Rawlins. Ingredients of instrumental meaning. *Journal of Semantics*, 34 (3):507–537, 2017. doi: 10.1093/jos/ffx003.

Barry Schein. *Plurals and Events.* MIT Press, Cambridge, MA, 1993.

Barry Schein. *'And': Conjunction Reduction Redux.* MIT Press, Cambridge, MA, 2017.

Barry Schein. Negation in event semantics. In Viviane Déprez and M. Teresa Espinal, editors, *The Oxford Handbook of Negation*, chapter 18, pages 301–332. Oxford University Press, Oxford, 2020.

Roger Schwarzschild. Pure event semantics. *Philosophical Perspectives*, 38(1):54–88, 2025.

Richard Sharvy. A more general theory of definite descriptions. *The Philosophical Review*, 89(4): 607–624, 1980.

Elizabeth S. Spelke and Katherine D. Kinzler. Core knowledge. *Developmental Science*, 10(1): 89–96, 2007. doi: 10.1111/j.1467-7687.2007.00569.x.

Brent Strickland. Language reflects "core" cognition: A new theory about the origin of cross-linguistic regularities. *Cognitive Science*, 41(1):70–101, 2017. doi: 10.1111/cogs.12332.

Leonard Talmy. Force dynamics in language and cognition. *Cognitive Science*, 12(1):49–100, 1988. doi: 10.1207/s15516709cog1201_2.

Carol Tenny and James Pustejovsky. *Events as Grammatical Objects: The Converging Perspectives of Lexical Semantics and Syntax*. CSLI Publications, Stanford, 2000.

Judith Jarvis Thomson. The time of a killing. *Journal of Philosophy*, 68(5):115–132, 1971.

Judith Jarvis Thomson. *Acts and Other Events*. Cornell University Press, Ithaca, 1977.

Judith Jarvis Thomson. Parthood and identity across time. *Journal of Philosophy*, 80(4):201–220, 1983. doi: 10.2307/2026004.

John C. Trueswell, Michael K. Tanenhaus, and Susan M. Garnsey. Semantic influences on parsing: Use of thematic role information in syntactic ambiguity resolution. *Journal of Memory and Language*, 33(3):285–318, 1994.

Robert Truswell. *The Oxford Handbook of Event Structure*. Oxford University Press, Oxford, 2019.

Ercenur Ünal, Frances Wilson, John C. Trueswell, and Anna Papafragou. Asymmetries in encoding event roles: Evidence from language and cognition. *Cognition*, 2024.

Alexis Wellwood. On the semantics of comparison across categories. *Linguistics and Philosophy*, 38(1):67–101, 2015.

Alexis Wellwood. *The Meaning of More*. Oxford University Press, Oxford, 2019.

Alexis Wellwood. Interpreting degree semantics. *Frontiers in Psychology*, 10:1–14, 2020.

Alexis Wellwood. Framing events in the logic of verbal modification. In *Proceedings of SALT 32*, pages 294–313, 2022.

Alexis Wellwood. Linguistic meaning needs stable concepts. *Philosophical Psychology*, submitted.

Alexis Wellwood, Angela Xiaoxue He, Jeffrey Lidz, and Alexander Williams. Participant structure in event perception. *University of Pennsylvania Working Papers in Linguistics*, 21(1):1–9, 2015.

Alexis Wellwood, Susan J. Hespos, and Lance Rips. How similar are objects and events? *Acta Linguistica Academica*, 15(2–3):473–501, 2018a.

Alexis Wellwood, Susan J. Hespos, and Lance Rips. The object:substance :: Event:process analogy. In Tania Lombrozo, Joshua Knobe, and Shaun Nichols, editors, *Oxford Studies in Experimental Philosophy*, volume 2, pages 183–212. Oxford University Press, Oxford, 2018b.

Alexander Williams. *Complex Causatives and Verbal Valence*. PhD thesis, University of Pennsylvania, 2005.

Alexander Williams. Themes, cumulativity, and resultatives: Comments on Kratzer 2003. *Linguistic Inquiry*, 40(4):686–700, 2009. doi: 10.1162/ling.2009.40.4.686.

Alexander Williams. *Arguments in Syntax and Semantics*. Cambridge University Press, Cambridge, 2015.

Alexander Williams. Events in semantics. In Piotr Stalmaszczyk, editor, *The Cambridge Handbook of the Philosophy of Language*, chapter 20. Cambridge University Press, Cambridge, 2021. doi: 10.1017/9781108698283.021.

Amanda L. Woodward. Infants selectively encode the goal object of an actor's reach. *Cognition*, 69(1):1–34, 1998. doi: 10.1016/S0010-0277(98)00058-4.

Susanne Wurmbrand and Magdalena Lohninger. An implicational universal in complementation: Theoretical insights and empirical progress. In Jutta M. Hartmann and Angelika Wöllstein, editors, *Propositional Arguments in Cross-Linguistic Research: Theoretical and Empirical Issues*, volume 84 of *Studien zur Deutschen Sprache*, pages 183–231. Gunter Narr Verlag, Tübingen, 2023.

Jeffrey M. Zacks and Barbara Tversky. Event structure in perception and conception. *Psychological Bulletin*, 127(1):3–21, 2001. doi: 10.1037/0033-2909.127.1.3.

Rolf A. Zwaan and Gabriel A. Radvansky. Situation models in language comprehension and memory. *Psychological Bulletin*, 123(2):162–185, 1998. doi: 10.1037/0033-2909.123.2.162.